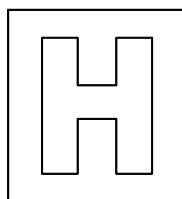


Candidate Name: _____

Class Adm No

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2013 Preliminary Examination II Pre-University 2

H1 Biology

8875/02

23 September 2013

2 hours

Additional Materials: Writing paper

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.

Write your Index number and name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions on the question booklet.

Section B

Answer **one** question on writing paper.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question. At the end of the examination, fasten all your work securely together.

For Examiner's Use	
1	
2	
3	
4	
5	
6 or 7	
Total	

This question paper consists of 12 printed pages.

[Turn over

Section A

Answer **all** questions in this section.

1. A scientist carried out an investigation to determine the effect of a newly discovered compound Y on the rate of an enzyme catalysed reaction. The results of a control experiment without Y added and an experiment with Y added are shown in the graph in Fig. 1.1.

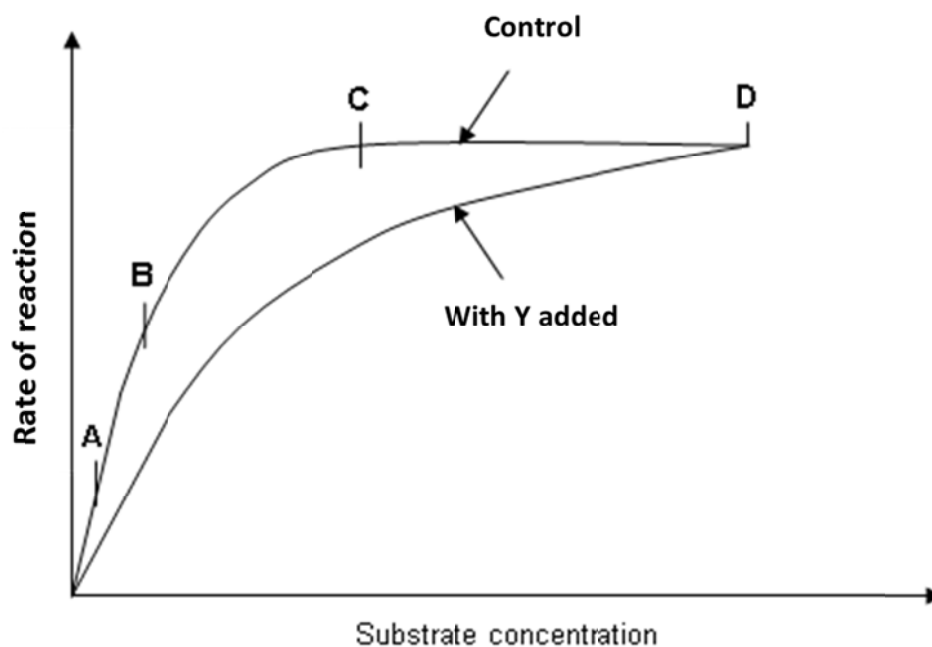


Fig. 1.1

- (a) With reference to Fig. 1.1, suggest the identity of Y and explain the reasons for the difference between the two graphs between points **A** and **B**.

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- (b)** Explain why the two graphs eventually reach the same maximum at point **D**.

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..... [2]

- (c)** Suggest two mechanisms by which an enzyme can lower activation energy of a reaction.

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..... [2]

[Total: 7]

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- The diagram illustrates the inner mitochondrial membrane, which separates the intermembrane space (top) from the mitochondrial matrix (bottom). The electron transport chain (ETC) is represented by a series of complexes in the membrane, with protons (H^+) being pumped from the matrix into the intermembrane space. ATP synthase is also shown, allowing protons to flow back into the matrix. FCCP (FCCP) is shown as a small molecule that can cross the membrane, and its addition is indicated by a double-headed arrow between FCCP and FCCP + H^+ . The intermembrane space is labeled "Intermembrane space" and the matrix is labeled "Mitochondrial matrix".

A student added a small quantity of FCCP to cells isolated from rat liver. He observed that this addition resulted in an increase in oxygen consumption by the cells, however, ATP production decreased.

- [4

Fluoroacetate is an inhibitor of aconitase, one of the enzymes involved in the Krebs cycle.

- (b)** If fluoroacetate was added to cells instead of FCCP, suggest the effect of this addition on the process of oxidative phosphorylation.

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- (c)** Describe the role of NAD in aerobic respiration.

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..... [4]

[Total: 11]

3. Rickets is a disease that often affects children, causing softening and weakening of bones. One of the causes of this disorder is caused by a mutation in an allele found on the X chromosome.

Draw two genetic diagrams to show how:

- two parents who do not have the disease could have a son with rickets.
- a man with rickets and a woman without rickets could have a daughter with rickets.

[6]

[Total: 6]

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4. Fig. 4.1 shows two sections from an onion root tip undergoing mitosis.

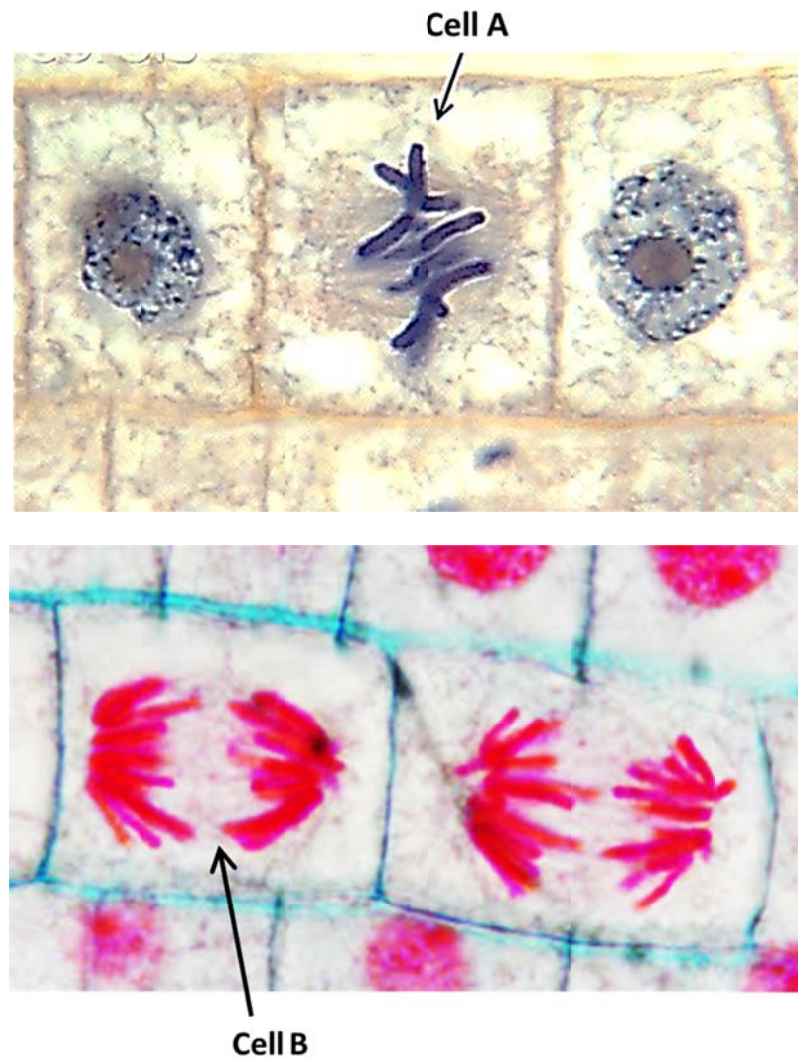


Fig. 4.1

- (a) State the stage of nuclear division that cells A and B are undergoing.

Cell A:

Cell B: [2]

- (b)** Describe what occurs in the cell during the two stages that you have identified in **(a)**.

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- (c)** Explain the significance of mitosis in cell division.

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[Total: 8]

5. A zombie outbreak is currently threatening the survival of humanity. Scientists have discovered that the cause of the zombie outbreak is a result of a 35 base pair deletion in a gene named *zomb*, coding for protein carrier Z. As a result of this mutation, a new restriction site for the restriction enzyme BamHI is introduced. Fig. 5.1 shows the restriction map for normal *zomb* and mutant *zomb*.

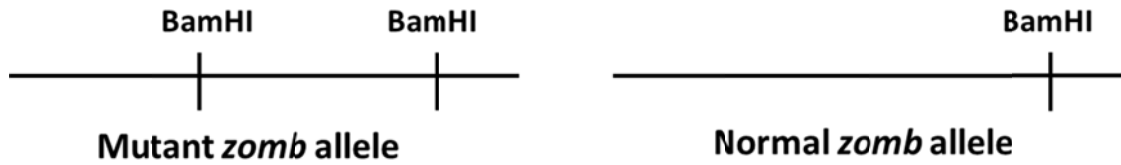
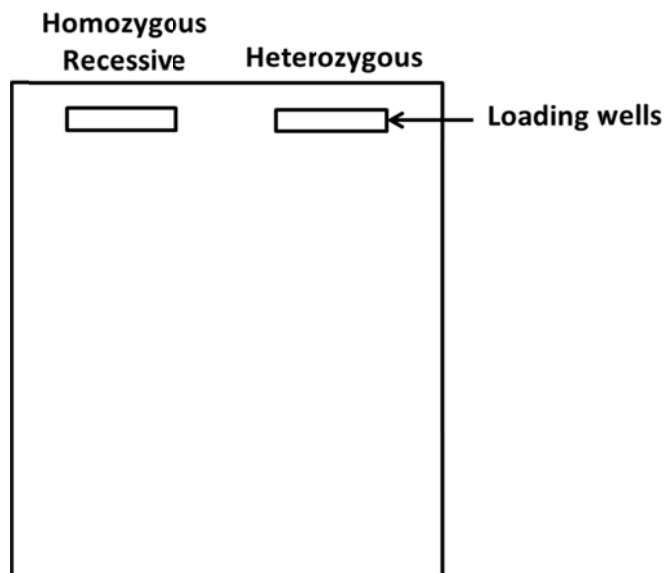


Fig. 5.1

The DNA samples of an individual who is homozygous recessive for the mutant *zomb* allele and an individual who is heterozygous for the *zomb* gene locus are subjected to restriction digestion by BamHI. The digested samples are then analysed via gel electrophoresis.

- (a) Draw the restriction pattern of these two individuals when their restriction fragments are analysed via gel electrophoresis.



[2]

- (b) Explain how restriction fragments are visualised during gel electrophoresis.

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A bite from a zombie is usually sufficient to convert a normal individual into a zombie. However, a study done by the World Health Organisation (WHO) showed that the time lapse between the moment an individual is bitten to the time he/she is converted into a zombie varies greatly within a population. Based on these observations the WHO constructed a 5-point scale to express this variation. The scale is shown in Fig. 5.2. The variation observed was found to be a result of different allele variants of a particular gene X.

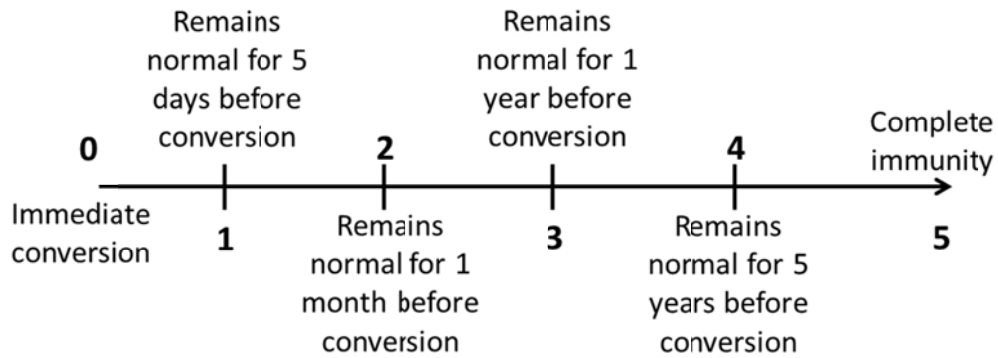


Fig. 5.2

- (c) State what type of natural selection would act on the population as a result of the zombie outbreak.

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- (d) With reference to the scenario given, use Darwin's theory of natural selection to explain how the allele frequencies of gene X in the population would change over time.

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[Total: 8]

Section B**Answer EITHER 6 OR 7.**

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

EITHER

6. (a) Using a named example, explain how the structure of a fibrous protein is related to its function. [8]
- (b) Describe the pathway taken by insulin from the time it is synthesised till it is secreted from the cell. [6]
- (c) Discuss the functions of triglycerides in animals. [6]
- [Total: 20]

OR

7. (a) Using a named example, explain how a change in DNA sequence may affect the phenotype of an organism. [4]
- (b) Discuss how gene or chromosomal mutations may result in cancer. [8]
- (c) Describe how the information on DNA may be used to synthesise a mature mRNA. [8]
- [Total: 20]